JAYASURYA T APPU

Senior Node.js / Backend Engineer
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PROFESSIONAL SUMMARY

Senior Node.js developer with 4+ years of experience building production backend systems: REST APIs, Redis Streams async pipelines, authentication and RBAC, and payment and messaging integrations. Supported ~1,000 requests/sec in production and cut API response times by 30–50% through MongoDB tuning. Seeking Node.js, backend, platform, and forward-deployed engineering roles.

KEY ACHIEVEMENTS

- Supported ~1,000 requests/sec in production on Node.js backend services.
- Contributed to an ~80% reduction in service cancellations through a Worker & Staff Analytics module.
- Cut API response times by 30–50% through MongoDB schema, indexing, and query optimization.
- Reduced manual HR effort by ~70–80% with a payroll automation system.
- Improved processing performance by ~20–40% by refactoring legacy backend modules.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, SQL, C
Backend: Node.js, Express.js, REST APIs, Authentication & Authorization, RBAC, Session Management, Redis Streams, Background Workers, Finite State Machine (FSM) Workflows
Databases: MongoDB, MySQL, Redis
Infrastructure: Docker, Docker Compose, Nginx, gRPC (project experience), Linux, Git, AWS (Basics)
Frontend: React.js, Angular, HTML5, CSS3
AI & Other: Python, FastAPI, RAG, Vector Embeddings, ChromaDB, Postman
Core Competencies: Backend Architecture, API Design, Performance Tuning, Asynchronous Processing, Third-Party Integrations, Code Reviews, Production Debugging

PROFESSIONAL EXPERIENCE

Spericorn — Senior Node.js Developer Aug 2026 – Present

- Developing and maintaining Node.js backend services.

Carclenx (Pexa) — Software Engineer Jul 2025 – Aug 2026

- Built Node.js REST APIs for production web and mobile applications, supporting ~1,000 requests/sec.
- Built asynchronous pipelines with Redis Streams and background workers, moving long-running jobs out of the request cycle.
- Built a Worker & Staff Analytics module with task-allocation logic, contributing to an ~80% reduction in service cancellations.
- Implemented authentication, session management, and Role-Based Access Control (RBAC) across user roles and admin workflows.
- Built FSM-driven workflows for conversational state and business processes, with messaging and payment gateway integrations using validation, retries, and fault-tolerant error handling.
- Built a payroll automation system for ~200 employees, reducing manual HR effort by ~70–80%.
- Tuned MongoDB schemas, indexes, aggregation pipelines, cursor-based pagination, and geospatial queries, cutting API response times by 30–50%.
- Defined API contracts with the engineering team and reviewed backend architecture, following Clean Architecture principles.

Cordova Educational Solutions — Software Developer Oct 2024 – Jun 2025

- Developed and maintained backend services and REST APIs for a Learning Management System (LMS).
- Refactored tightly coupled legacy modules into reusable components, improving performance by ~20–40%.
- Worked with product stakeholders to translate business requirements into production-ready features.
- Investigated production issues through debugging and root cause analysis, and built React.js features.

Quest Innovative Solution — Junior Software Engineer Dec 2022 – Sep 2024

- Built and maintained Node.js/Express.js REST APIs for multiple client applications, including booking systems.
- Integrated third-party APIs and external services across client projects.
- Optimized relational database queries and backend logic for performance and data consistency.
- Built responsive React.js interfaces integrated with backend APIs.

Carclenx (Pexa) — Full Stack Developer Jan 2022 – Jun 2022

- Built RESTful APIs with Node.js/Express.js and integrated third-party services.
- Built Angular admin dashboards integrated with backend APIs.

PROJECTS

Distributed Commerce Platform | *Node.js, Express.js, gRPC, Redis, MongoDB, Docker, Nginx* | GitHub

- Microservices e-commerce platform with independently deployable services and gRPC inter-service communication.
- Redis caching, Nginx API gateway, and Docker Compose for production-like local deployment.

Enterprise Knowledge Platform | *Python, FastAPI, ChromaDB, Sentence Transformers, LLMs* | GitHub

- RAG platform for semantic search and AI-powered question answering over document collections.
- Document ingestion pipeline (chunking, embeddings, vector indexing) with FastAPI REST endpoints.

Forge – Build Automation Tool | *C, GCC, Windows API* | GitHub

- Command-line tool in C that scans project directories and generates GCC compilation commands.

EDUCATION

B.Tech, Computer Science & Engineering — MG College of Engineering 2017 – 2021
Degree not completed